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Energy efficient clustering and routing protocol based on quantum particle swarm optimization and fuzzy logic for wireless sensor networks

Huangshui Hu¹, Xinji Fan¹ & Chuhan Wang^{2✉}

Clustering and routing protocols play a pivotal role in reducing energy consumption and extending the lifespan of wireless sensor networks. However, optimizing energy efficiency to maximize network longevity remains a primary challenge for these protocols. This paper introduces QPSOFL, a clustering and routing protocol that integrates quantum particle swarm optimization and a fuzzy logic system to enhance energy efficiency and prolong network lifespan. QPSOFL employs an enhanced quantum particle swarm optimization algorithm to select optimal cluster heads, utilizing Sobol sequences for population diversification during initialization. Additionally, it incorporates Lévy flight and Gaussian perturbation-based position updates to prevent trapping in local optima. Benchmark experiments validate QPSOFL's efficacy compared to Harris Hawks Optimization (HHO), Grey Wolf Optimization (GWO), Particle Swarm Optimization (PSO), and Quantum Particle Swarm Optimization (QPSO), focusing on accuracy, search capability, and convergence speed. Within QPSOFL, a fuzzy logic system determines the best next-hop cluster head based on descriptors such as residual energy, energy deviation, and relay distance. Extensive simulations compare QPSOFL's performance in terms of network lifetime, throughput, energy consumption, and scalability against existing protocols E-FUCA, IHHO-F, F-GWO, and FLPSOC, demonstrating its superior performance over these counterparts.

Keywords Wireless sensor networks, Quantum particle swarm optimization, Energy efficiency, Clustering and routing, Fuzzy logic

Abbreviations

WSNs	Wireless sensor networks
CH	Cluster head
CM	Cluster members
QPSO	Quantum particle swarm optimization
PSO	Particle swarm optimization
$E_{res}(CH_i)$	The remaining energy of candidate relay i of CH j
$E_{dev}(CH_i)$	The deviation of energy consumption of candidate CH i
$D_{ry}(CH_i)$	The sum of distances d_r and d_p
E_{elec}	The electronics energy consumption used for transmitting or receiving 1-bit data
E_0	The initial energy of node i
$\varepsilon_{fs}, \varepsilon_{mp}$	The amplifier coefficients used for free space and multi-path model
d_0	The distance threshold for the model determination

Wireless Sensor Networks (WSNs) serve as a fundamental data acquisition technology for the Internet of Things (IoT), finding extensive applications in environment monitoring¹, target tracking², intelligent transportation³, healthcare⁴, among others. In WSNs, numerous sensor nodes collaborate to fulfill specific tasks, constrained

¹College of Computer Science and Engineering, Changchun University of Technology, Changchun, China. ²College of Computer Science and Technology, Changchun Normal University, Changchun, China. ✉email: chuhangw@163.com

by limited energy, computation, storage, and communication capabilities. Typically, a Base Station (BS), which remains unconstrained in resources, manages command issuance, data collection, and processing. Energy efficiency emerges as critical for extending network lifetime due to challenges like hostile environments, irreplaceable or non-rechargeable energy sources, and dynamic topology. Clustering and routing protocols have emerged as effective solutions⁵. These protocols organize nodes into clusters, each comprising a Cluster Head (CH) and Cluster Members (CMs). CHs, distinct from regular CMs, undertake responsibilities such as data reception, aggregation, and transfer, in addition to information sensing and transmission. Consequently, CHs are prone to premature depletion due to rapid energy consumption. Optimal CH selection, along with CH rotation or burden sharing via backup CHs, represents common strategies to mitigate this issue^{6,7}. Once CHs are selected, ordinary nodes establish communication with CHs to form clusters. Furthermore, intra-cluster data transmission is managed using a Time Division Multiple Access (TDMA) mechanism to conserve energy by avoiding interference and collisions. Subsequently, a route determination method identifies the optimal next-hop CH for each CH, ensuring reliable data transmission from source nodes to the BS. Since the inception of the pioneering clustering and routing protocol Low Energy Adaptive Clustering Hierarchy (LEACH), numerous enhanced protocols have been proposed from diverse perspectives, achieving notable performance improvements⁸.

Random selection of Cluster Heads (CHs) based on probability, disregard for residual energy during CH selection, and single-hop data transmission to the BS are the primary drawbacks of LEACH^{8,9}. Consequently, weight-based protocols considering multiple parameters and enabling multi-hop communication have been proposed to address these issues^{8,10,11}. However, clustering and routing remain typical NP-hard problems¹², challenging traditional methods to find optimal solutions. Therefore, intelligent computing approaches such as Genetic Algorithm (GA)¹², Grey Wolf Optimization¹³, Fire Hawk Optimizer (FHO)¹⁴, among others^{5,8}, have been introduced to achieve optimal or approximate solutions for clustering and routing optimization⁵. Despite their demonstrated performance, these methods struggle with uncertainties and interferences inherent in clustering and routing processes. Hence, fuzzy logic is increasingly employed to mitigate these challenges, leveraging expert knowledge bases to enhance decision-making capabilities under uncertain and chaotic environments. Fuzzy logic has found wide application in WSNs for clustering¹⁵, routing¹⁶, and their integration^{17,18}. Moreover, fuzzy logic is synergistically combined with other technologies such as ant colony algorithms¹⁹, butterfly optimization algorithms⁶, tunicate swarm algorithms²⁰, and notably, particle swarm optimization²¹, effectively integrating their strengths to enhance overall network performance. Particle swarm optimization, a popular nature-inspired heuristic, is extensively used for solving single and multi-objective optimization problems in domains like feature selection, wireless communications, and image processing²².

Particle Swarm Optimization is inspired by observing the social behavior of birds flocking and fish schooling, where particles represent candidate solutions searching the solution space across given dimensions to find the optimal solution. PSO is often preferred over other nature-inspired methods for solving NP-hard problems due to its ease of implementation, rapid convergence, and minimal need for parameter adjustments²³. Consequently, PSO finds application in various tasks within WSNs such as target tracking², sink mobility²⁴, clustering²³, routing²¹, and combined clustering and routing²⁵. Despite its ability to balance exploration and exploitation for good network performance, PSO has notable weaknesses including premature convergence, challenges in velocity updating, and parameter tuning difficulties. To enhance optimization performance, several new PSO variants have been developed, with Quantum Particle Swarm Optimization emerging as one of the most effective. QPSO exhibits greater exploratory tendencies to avoid local optima, features a simpler update mechanism adaptable to diverse problems, and demonstrates superior convergence behavior leading to highly optimal solutions. Extensive testing confirms QPSO's superiority over classical PSO, its variants, and other intelligent computation algorithms²⁶, making it widely applicable in suspension optimization²⁷, multi-task allocation²⁸, and other domains. However, QPSO's efficiency in achieving global optimum solutions is hindered by its low diversity and limited search capabilities. Consequently, numerous improvements to QPSO aim to strike a balance between local and global search capabilities while mitigating premature convergence issues²⁹. Despite its strengths, QPSO has yet to be extensively applied to clustering and routing protocols in WSNs. Hybridizing PSO with other methods presents a promising research direction for enhancing clustering and routing, such as integrating with fuzzy logic²¹, genetic algorithms³⁰, or lion swarm optimization³¹. This approach aims to leverage the strengths of different algorithms, combining them to overcome individual drawbacks and achieve superior results in solving complex problems.

In this paper, a novel clustering and routing protocol hybridizing quantum particle swarm optimization with fuzzy logic called QPSOFL is proposed to improve the network energy efficiency and enhance the network lifetime. In QPSOFL, QPSO is used to select the optimal CHs for clusters formation, and fuzzy logic system is applied to find the best relay CHs for data transmission. The contributions of the proposed protocol are summarized as follows:

1. Energy-efficient clusters are determined using a quantum particle swarm optimization algorithm, employing Sobol sequences, Lévy flights, and Gaussian perturbation to broaden the search space, enhance convergence speed, and mitigate local optima trapping;
2. Energy-efficient and balanced relay CHs are achieved through a Mamdani fuzzy inference system. Parameters such as residual energy, energy deviation, and relay distance are integrated to guide decision-making based on fuzzy rules derived from expert experience;
3. Function test-bench and simulations are conducted to assess the performance of the enhanced QPSO algorithm and QPSOFL protocol respectively, compared to existing approaches. The results confirm the effectiveness of the proposed protocol.

The rest of this paper is organized as follows. The literature review of clustering and routing protocols is presented in Section "Related works". The improved quantum particle swarm optimization algorithm is addressed in Section "Improved quantum particle swarm optimization algorithm". In Section "Propose protocol", the proposed protocol QPSOFL is introduced in detail. Performance evaluation is outlined in Section "Simulation results", and finally, Section "Conclusion" concludes the paper.

Related works

Clustering and routing protocols represent a significant focus of current research in wireless sensor networks, aiming to improve network efficiency, extend sensor node lifespan, and optimize data transmission. This section categorizes recent academic research into three distinct segments, highlighting advancements across various domains. Firstly, we review clustering and routing approaches employing fuzzy logic. Secondly, we explore the application of intelligent computing techniques, specifically Particle Swarm Optimization algorithms, to enhance the efficiency of clustering and routing operations. Lastly, we examine approaches that integrate fuzzy logic with intelligent algorithms, effectively addressing inherent challenges in WSNs.

Traditional clustering and routing methods

The LEACH³² protocol is a classical clustering and routing protocol designed for wireless sensor networks with the primary goal of optimizing network resource utilization. In its operation, the sensor nodes within the network undergo an initial clustering process, with each cluster autonomously selecting a CH node. These elected CH nodes play a pivotal role in centralized data aggregation and transmission, effectively mitigating energy consumption and thereby extending the overall network lifespan. However, the reliance on a random process for selecting CH nodes introduces a potential concern where specific nodes may frequently serve as CHs, resulting in an uneven distribution of energy consumption, which exerts a detrimental impact on the overall network lifetime. Therefore, a fuzzy logic-based clustering protocol is proposed in³³. Unlike LEACH, a fuzzy inference system is used in the formation of clusters to calculate the chance value of all nodes to be a CH based on the residual energy of the node, the distance to the BS, and the number of neighboring nodes. CHs selected based on chance value clearly perform better than randomly selected CHs, as verified in simulation. Nonetheless, the single-hop clustering protocol is insufficient for data transmission over a larger network area solely by selecting the CH nodes. A multi-directional routing protocol incorporating fuzzy clustering is proposed in³⁴. Initially, the CH node is selected based on the residual energy of the node and the distance from the BS, combined with a fuzzy logic system. Then, multiple paths that may exist between the CH node and the BS are identified, and the path with the fewest number of hops is selected as the routing path for data transmission, proven effective in extending the network lifetime through simulation results. The above protocols may lead to premature death of CH nodes close to the BS due to unequal energy consumption. Therefore, two fuzzy inequality clustering and routing protocols, FUCA and E-FUCA, are proposed in³⁵ and³⁶ respectively, where E-FUCA³⁶ is an enhancement of FUCA³⁵. The proposed unequal clustering protocol FUCA aims to improve the survival time of the entire network. Unlike other fuzzy-based clustering algorithms, FUCA has two output variables, "Competition Radius" and "Rank," and uses three inputs for its computation, namely "Residual Energy," "Node density," and "Distance to BS." Computation of competition radius and ranking using predefined fuzzy "if-then" rules allows the competition radius and ranking of each initial CH to change dynamically. This reduces the competitive radius of CHs while decreasing energy, thereby forming unequal clusters. The output linguistic variables are defuzzified using a region-centric approach to obtain a clear value. After completing the CH election, non-CH nodes join the nearest CH to form a cluster. After cluster formation, non-CH nodes transmit sensed data to their respective CHs, and each CH summarizes all the received data and sends it to the receiver. In contrast, E-FUCA does not consider node density but focuses on the average distance to the communication node, as node density does not provide a complete picture of the energy consumption of the CH. Choosing the average distance offers a clear picture of the communication cost borne by the CH. In FUCA, non-CH nodes make a greedy decision during cluster formation, selecting the nearest CH without considering the existing load of the CH. In E-FUCA, non-CH nodes intelligently select their CHs based on the CH's rank, proximity to the CH, and the number of nodes within the cluster radius obtained during CH selection. Additionally, E-FUCA selects its CHs based on the Next hop rank, Nearness to the next-hop, and the Distance reduced to BS, further enhancing energy efficiency.

Intelligent computing based clustering and routing methods

Traditional clustering-based routing algorithms facilitate fault tolerance, load balancing, and reliable communication. However, they compromise the lifetime of CHs. The rising popularity of intelligent computing algorithms has introduced novel concepts to clustering and routing protocols in WSNs. In³⁷, an energy-efficient CH selection method was introduced. This method employs an enhanced GWO algorithm and considers various parameters, including convergence distance, residual energy, balancing factor, and average intra-cluster distance when selecting CHs, so as to improve the lifetime of the WSN. The fitness function for CH selection is designed based on factors such as residual energy and the Euclidean distance between nodes and the BS. This design aims to achieve uniform CH selection in each round, thereby promoting energy balance and extending the network lifetime. Similarly, another widely adopted population-based smart optimization algorithm is PSO. In CASIC-PSO³⁸, a method utilizes PSO to select suitable sensor nodes as CHs with the aim of prolonging the network lifetime. Initially, CASIC-PSO performs a random initialization of each particle, setting its velocity and position by mapping particle encoding to the coordinates of sensor nodes. Subsequently, a fitness function is established to compute the fitness values of particles, taking into account sensor node's remaining energy and transmission energy consumption between sensor nodes and the BS. The personal best fitness value for each particle and the global best fitness value for personal best values are then determined. Finally, the particles update their

coordinates to achieve the global best solution based on position and velocity vectors until the predefined number of iterations is reached. Additionally, the acceleration coefficients are adaptively adjusted to enhance the search quality with respect to the iteration number. The PDU-SLNO algorithm is introduced in³⁹, which integrates sea lion hunting behavior with the particle swarm optimization algorithm. Specifically, the PDU-SLNO algorithm mimics the hunting strategies of sea lions, encompassing steps such as chasing, tracking, following, surrounding, and attacking prey. Concurrently, it incorporates PSO's global and local search methods to determine the optimal position. During implementation, each particle is assigned a speed and random position, navigating through the search space. In each iteration, particles converge towards their optimal positions. By combining sea lion behavior with PSO, the PDU-SLNO algorithm effectively addresses the challenge of cluster header selection. In IPSO-GWO⁴⁰, another method is introduced to select optimal CHs and find routing paths within the network. Primarily, CH selection is viewed as an optimization problem in IPSO-GWO, which is addressed using an enhanced PSO named as Improved Particle Swarm Optimization (IPSO). Its objectives include maximizing the ratio of average residual energy of CHs to that of non-CH nodes, as well as maximizing the ratio of the maximum distance among non-CH nodes and the BS to the average distance among BS and CHs. Additionally, the inertial weight is adaptively adjusted to prevent getting stuck in local optima. It should be noted that random particle initialization may slow down the convergence process. After the selection of CHs by IPSO, non-CH nodes become part of clusters through interactions with the nearest CHs. Subsequently, GWO is applied to determine the optimal paths, aiming to minimize distance traversal and hop count. The best solution is achieved by updating the speed and location of the wolves. It's worth noting that residual energy is not taken into account in path selection. Simulation results demonstrate that IPSO-GWO can provide maximal energy efficiency while extending the network lifetime. In⁴¹, PSO-EEC is introduced as a method for selecting CHs and relay nodes to enhance network lifetime and performance. PSO-EEC follows operations similar to traditional PSO to designate the most efficient nodes as CHs and optimal CHs as relays. To achieve this, fitness functions are created to assess the particles, and the particles adjust their position and velocity based on the local and global best to attain the ultimate global optimal solution. The fitness function for clustering takes into account parameters such as the ratio of nodes' initial energy to residual energy, the distance between cluster member nodes and the Cluster Head, and node degree to select the most suitable nodes for the role of CH. Simultaneously, the fitness function for routing considers the residual energy of CHs and the distance between CHs and the BS to identify the optimal relay nodes for transmitting data collected by the CHs to the BS. Experimental results demonstrate that PSO-EEC has the potential to enhance network lifetime and performance. Likewise, in⁴², a multi-hop clustering routing algorithm called EBPSO is introduced, which is also based on PSO. In the initial phase of CH selection, the most suitable CH is chosen by taking into account the energy of the sensor nodes, the distance between nodes within the cluster, and the distance between the CH and the BS. In the subsequent data transmission phase, a relay node selection mechanism is introduced, favoring nodes with higher remaining energy and proximity to the BS for selection as relay nodes. Furthermore, to mitigate energy consumption and distinguish from the majority of cluster routing protocols employing a Round-Based Policy (RBP), the paper establishes an appropriate energy threshold. Re-clustering is only conducted when the energy of cluster nodes falls below this threshold to prevent excessive energy consumption resulting from frequent clustering.

Hybrid clustering and routing methods

During the last decades, numerous protocols that blend conventional approaches with intelligent computing algorithms have been proposed to improve the network performance. To enhance network longevity, a fuzzy-based Improved Harris's Hawk Optimization Algorithm (IHHO-F) is present in⁴³. This algorithm aims to identify the optimal set of CHs by navigating both the exploration and exploitation phases. This strategy employs differential evolution to converge the solution set, generating P^* . In each round, the node closest to P^* represents the solution set containing CHs. Subsequently, fuzzy logic is employed to select the optimal CHs. Likewise, solutions are grouped into clusters using a fuzzy logic system to identify the optimal cluster head set based on three parameters: the current energy of CH nodes, the distance from the cluster head to the convergence point, and the number of neighboring nodes of the CHs. A node is designated as a Super Cluster Head (SCH) if it possesses higher residual energy, closer proximity to the BS, and a greater density of neighboring nodes. During the data transmission phase, the CH collects data from member nodes. If the number of hops is 1, it directly transmits the data to the BS. Otherwise, it routes the data through the intermediate CH. Subsequently, the SCH employs a time-sharing method to gather data from the respective CH and forwards it to the BS. In⁴⁴, a method combining GWO and a fuzzy system is proposed to conserve energy and prolong the sensor network's lifetime. This approach considers node centrality, neighborhood overlap, and residual energy within the wireless sensor network, and it selects the node with higher energy and closer proximity to the BS to serve as the CH. Moreover, an energy-efficient opportunistic routing scheme is utilized to achieve the goal of minimizing network energy consumption. In⁴⁵, a fuzzy-based energy-efficient routing protocol is introduced. The protocol employs a PSO algorithm for clustering and a fuzzy logic system for data transmission. In PSO, it employs a fitness function that incorporates particle energy, inter-particle distance, and the number of clusters reachable by particles to select the best CHs. Additionally, four inputs battery energy, node density, average Signal-to-Noise Ratio (SNR) to BS distance, and communication quality are considered to choose the next hop CH. Simulation results demonstrate the effective enhancement of network lifetime by this clustering and routing protocol. In contrast, an energy-efficient clustering and routing approach based on fuzzy logic and PSO (FLPSOC) is proposed in⁴⁶, which employs a fuzzy logic system for initial CH selection. Input parameters include the ratio of initial to residual energy, the number of neighbors, and the distance between neighbors are considered. Subsequently, the PSO algorithm with a fitness function based on the residual energy of CHs and the distance between CHs and the BS is used to find the optimal

relay CHs. Simulations validate its effectiveness in terms of network lifespan, throughput, stability period, cluster head count and so on. The pairs between the mentioned protocols are shown in Table 1.

Improved quantum particle swarm optimization algorithm

As one of the most significant optimization algorithms, PSO has been proved its effectiveness and potential in various fields of optimization. The particles cooperate to search the best solution through learning which guides the population direction with consistency. Moreover, the knowledge of itself and its neighbors for learning is updated by calculating the quality of particles through a fitness function, which is represented by local best and global best. Hence, the adaptability degree and search capability of PSO are maximized and strengthened. Usually, In PSO, a particle in d -dimension space describes a potential solution of the optimization problem, which can be expressed as a position vector in Eq. (1).

$$X_i = [x_{i1}, x_{i2}, \dots, x_{id}] \tag{1}$$

A velocity vector with the same dimension is used to adjust the positions of the particle according to the local best and global best, then the new positions of the particle can be obtained by Eq. (2).

$$\begin{aligned} x_{id}^{t+1} &= x_{id}^t + v_{id}^{t+1} \\ v_{id}^{t+1} &= \omega \times v_{id}^t + c_1 rand() (pBest_i - x_{id}^t) + c_2 rand() (gBest - x_{id}^t) \end{aligned} \tag{2}$$

where t is the current iteration, i means the i th particle, $c_1 = c_2 = 2$ are two non-negative constants called acceleration factors³¹, $pBest_i$ and $gBest$ denote the local best and global best, respectively. Continuously learning from the previous experiences, the global best solution will be reached in the end. Although PSO is successful in most of optimization problems, its limitations of time consuming, convergence rate, and local optima similar to other swarm intelligent algorithm still deteriate the overall performance. Therefore, an improved quantum particle swarm optimization algorithm GQPSO combined Sobol sequence based population initialization with Lévy flight and Gaussian perturbation based position updating is proposed to prevent the shortcomings of the traditional PSO. In GQPSO, quantum particle with d -dimension $P_i = (q_{i,1}, q_{i,2}, \dots, q_{i,d}) (1 \leq i \leq N_p)$ is used to represent the solution of optimization problems, and N_p is the size of population. Different from random population initialization, Sobol sequence is adopted to generate the initial population, which is a low differentiation sequence generated by deterministic quasi-random numbers, so as to search the whole space for fast convergence and high accuracy. Then, the component of a particle can be expressed in Eq. (3).

$$q_{i,j} = lb + S_n \times (ub - lb) \tag{3}$$

where lb, ub are the upper and lower bounds of the search space, and S_n represents the random number generated by the Sobol sequence. Moreover, Lévy flight is utilized to expand the search space and improve the global searching ability so as to avoid being caught in local optimal. Hence, the new state of the quantum particle is updated by Eq. (4).

$$\begin{cases} q_{i,j}^{t+1} = L(\lambda) [q_{i,j}^t - p_i^t] + \alpha \cdot |\overline{p_{BESTi}} - q_{i,j}^t| \cdot \ln \left[\frac{1}{u_{i,j}^t} \right] & \text{if } k \geq 0.5 \\ q_{i,j}^{t+1} = p_i^t - \alpha \cdot |\overline{p_{BESTi}} - q_{i,j}^t| \cdot \ln \left[\frac{1}{u_{i,j}^t} \right] & \text{if } k < 0.5 \end{cases} \tag{4}$$

Protocols	Considered algorithm	Main idea	Objectives
32	–	Random selects CHs	Optimizing network resource utilization
33	Fuzzy logic system	Fuzzy logic selects CHs	Optimizing network resource utilization
34	Fuzzy logic system	Fuzzy logic selects routes	Proven effective in extending the network lifetime
35	Fuzzy logic system	Fuzzy logic combines unequal clustering	Enhancing energy efficiency
36	Fuzzy logic system	Enhancements for 35	Enhancing energy efficiency
37	GWO	The fitness function of CH selection is designed to achieve uniform CHs selection in each round	Promoting energy balance and extending the network lifetime
38	PSO	The particle is encoded and the fitness function value is calculated	Prolonging the network lifetime
39	PSO + SLnO	Combine the advantages of the two algorithms to select CHs	Maximize network life cycle and energy efficiency
40	PSO + GWO	Combine the advantages of the two algorithms to select routes	Maximal energy efficiency while extending the network lifetime
41	PSO	Select CHs and route based on fitness function values	Enhance network lifetime and performance
42	PSO	Select CHs and route based on fitness function values	Enhance network lifetime and performance
43	Fuzzy logic system + HHO	HHO determines the optimal CHs set, fuzzy logic selects the optimal CHs, SCH collects the data and forwards it	Enhance network lifetime and performance
44	Fuzzy logic system + GWO	Combine fuzzy and intelligent algorithms to select CHs and routes	Minimizing network energy consumption
45	PSO + Fuzzy logic system	Fuzzy logic clustering and PSO routing	Enhance network lifetime and performance
46	Fuzzy logic system + PSO	PSO clustering and fuzzy logic routing	Enhance network lifetime and performance

Table 1. Comparison of protocols.

where $p_i^t = \text{rand}() \bullet pBest_i^t + (1 - \text{rand}()) \bullet gBest$, k is a random number uniformly ranging in $(0, 1)$, $u_{ij} \in \text{rand}(0,1)$, and α represents the contraction expansion coefficient that determines the convergence rate. Linear-descending strategy is adopted to control α as shown in Eq. (5).

$$\alpha = \omega_0(\omega_1 - \omega_0) \frac{\text{iter}_{\max} - \text{iter}}{\text{iter}_{\max}} \quad (5)$$

iter and $\text{iter}_{\max}$ are current and maximum iteration number, respectively. ω_1 and ω_0 are factors generally taken as 1.0 and 0.5²⁷. $\overline{pBest_i}$ denotes the average best position ($\overline{pBest_i} = \frac{1}{N_p} \sum_{i=1}^{N_p} pBest_i^t$) used to make the particles converge to the local attractor. $L(\lambda) = 0.01 * \frac{u * \delta}{|v|^{\frac{1}{\beta}}}$ is Lévy flight function in which standard deviation δ can be expressed as follows.

$$\delta = \left(\frac{\Gamma(1 + \beta) * \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) * \beta * 2\left(\frac{\beta-1}{2}\right)} \right)^{\frac{1}{\beta}} \quad (6)$$

u and v are two random parameters ranges between $(0, 1)$, and β is a constant with value 1.5. Moreover, if the local best keeps the same during two consecutive iterations, the Gaussian perturbation is applied to update the state of the quantum particle, which can be expressed in Eq. (7).

$$P_i^{t+1} = \text{Gaussian}(P_i^t, \sigma) \\ \sigma = \cos\left(\frac{\pi}{2} * \left(\frac{t}{T}\right)^2 * (P_i^t - P_j^{t*})\right) \quad (7)$$

where P_j^{t*} ($j \neq i$) represents any particle in the population. The step size of the Gaussian random walk is determined by initially applying a significant perturbation to the Cosine function during early iterations. Subsequently, this perturbation decreases rapidly in later stages, thereby balancing the algorithm's exploration and exploitation capabilities. The single-peak function assesses the algorithm's ability to locate a global optimum within the search space, while the multi-peak function evaluates its adaptability to complex search spaces containing multiple locally optimal solutions. The fixed-dimensional multi-peak problem scrutinizes the performance of optimization algorithms in managing intricate search spaces with numerous locally optimal solutions. To evaluate the effectiveness of GQPSO, we conducted experimental tests using established benchmark functions detailed in Table 2. In the table, D denotes the dimension of the function, $f_{\min}$ representing the global optimum. The first four functions are single-peaked, the next four are multi-peaked, and the last four are multi-peaked with fixed dimensions. All algorithms use a population size of 200 and run for 2000 iterations. The experiments were conducted using MATLAB R2022a on a Windows 10 operating system with 16 GB of RAM. To enhance the experiment's validity and reduce the impact of random events, we compute the best, average (aver), and standard

Function	D	Search space	$F_{\min}$
$F1(x) = \sum_{i=1}^n x_i^2$	30	$[-100, 100]^D$	0
$F2(x) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	30	$[-10, 10]^D$	0
$F3(x) = \max_i \{ x_i , 1 \leq i \leq n\}$	30	$[-100, 100]^D$	0
$F4(x) = \sum_{i=1}^n ([x_i + 0.5])^2$	30	$[-100, 100]^D$	0
$F5(x) = -20 \exp(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2}) - \exp(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i)) + 20 + e$	30	$[-32, 32]^D$	0
$F6(x) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos(\frac{x_i}{\sqrt{i}}) + 1$	30	$[-600, 600]^D$	0
$F7(x) = \frac{\pi}{n} \{10 \sin^2(\pi y_1) + \sum_{i=1}^{n-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] + (y_n - 1)^2\} + \sum_{i=1}^n u(x_i, 10, 100, 4)$ $y_i = 1 + \frac{x_i + 1}{4}, u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m x_i & > a \\ 0 & -a < x_i < a \\ k(-x_i - a)^m x_i & < -a \end{cases}$	30	$[-50, 50]^D$	0
$F8(x) = 0.1 \{\sin^2(3\pi x_1) + \sum_{i=1}^{n-1} (x_i - 1)^2 [1 + \sin^2(3\pi x_{i+1})] + (x_n - 1)^2 [1 + \sin^2(2\pi x_n)]\} + \sum_{i=1}^n u(x_i, 5, 100, 4)$	30	$[-50, 50]^D$	0
$F9(x) = -\sum_{i=1}^4 c_i \exp(-\sum_{j=1}^3 a_{ij}(x_j - p_{ij})^2)$	3	$[0, 1]^D$	-3.86278214782076
$F10(x) = -\sum_{i=1}^4 c_i \exp(-\sum_{j=1}^6 a_{ij}(x_j - p_{ij})^2)$	6	$[0, 1]^D$	-3.32199517158424
$F11(x) = -\sum_{i=1}^5 [(x - a_i)(x - a_i)^T + c_i]^{-1}$	4	$[0, 10]^D$	-10.153199679
$F12(x) = -\sum_{i=1}^7 [(x - a_i)(x - a_i)^T + c_i]^{-1}$	4	$[0, 10]^D$	-10.4029405667869

Table 2. Benchmark functions.

Function	Algorithm	Best	Aver	Std
F1(x)	GQPSO	0	0	0
	QPSO	2.317e-09	7.1628e-12	6.6454e-09
	PSO	2.066e-20	7.7283e-22	2.5511e-20
	GWO	1.7377e-203	2.8813e-205	0
	HHO	0	0	0
F2(x)	GQPSO	0	0	0
	QPSO	1.708e-07	7.6408e-09	1.7121e-07
	PSO	4.5173e-14	4.0527e-15	3.6407e-14
	GWO	4.9077e-116	1.5666e-116	3.4968e-116
	HHO	1.2371e-218	1.8706e-238	0
F3(x)	GQPSO	0	0	0
	QPSO	0.88002	0.41416	0.34044
	PSO	0.25096	0.093646	0.24279
	GWO	2.8477e-52	4.7405e-52	2.071e-54
	HHO	5.5199e-208	1.5109e-218	0
F4(x)	GQPSO	0.38123	0.01705	0.29339
	QPSO	2.9865e-10	5.8306e-12	4.1297e-10
	PSO	9.9216e-20	2.4982e-21	1.4107e-19
	GWO	0.069079	8.5435e-07	0.1124
	HHO	1.3221e-07	1.3376e-080	1.0428e-07
F5(x)	GQPSO	4.4409e-16	4.4409e-16	0
	QPSO	7.1242e-06	7.2752e-07	7.5818e-06
	PSO	4.0089e-11	1.3845e-11	2.9605e-11
	GWO	7.5495e-15	7.5495e-15	0
	HHO	4.4409e-16	4.4409e-16	0
F6(x)	GQPSO	0	0	0
	QPSO	0.013773	1.2634e-09	0.012161
	PSO	0.01198	0	0.022662
	GWO	0.0017423	0	0.0037185
	HHO	0	0	0
F7(x)	GQPSO	0.014685	0.0066633	0.006974
	QPSO	3.3664e-11	5.042e-12	5.7209e-11
	PSO	2.0336e-20	6.1081e-23	6.1599e-20
	GWO	0.003283	6.0822e-08	0.0046516
	HHO	1.6264e-08	9.955e-10	1.5948e-08
F8(x)	GQPSO	0.71186	0.43782	0.26227
	QPSO	0.0032962	1.1377e-10	0.0053074
	PSO	1.554e-20	1.2515e-22	1.7936e-20
	GWO	0.059758	8.1989e-07	0.070104
	HHO	9.7393e-08	2.4331e-09	1.2694e-07
F9(x)	GQPSO	− 3.8628	− 3.8628	2.6285e-07
	QPSO	− 3.8628	− 3.8628	9.3622e-16
	PSO	− 3.8628	− 3.8628	9.3622e-16
	GWO	− 3.8612	− 3.8628	0.0032924
	HHO	− 3.8628	− 3.8628	1.5326e-13
F10(x)	GQPSO	− 3.2506	− 3.322	0.061399
	QPSO	− 3.2388	− 3.322	0.057431
	PSO	− 3.2744	− 3.322	0.061396
	GWO	− 3.2312	− 3.322	0.065744
	HHO	− 3.2744	− 3.322	0.061409
F11(x)	GQPSO	− 10.1526	− 10.1532	0.00053545
	QPSO	− 9.1259	− 10.1532	2.1642
	PSO	− 8.3859	− 10.1532	2.9226
	GWO	− 9.1427	− 10.1532	2.1303
	HHO	− 5.565	− 10.1532	1.6121
F12(x)	GQPSO	− 10.4015	− 10.4029	0.0012519
Continued				

Function	Algorithm	Best	Aver	Std
	QPSO	− 9.5734	− 10.4029	2.4008
	PSO	− 7.2892	− 10.4029	3.3925
	GWO	− 10.4029	− 10.4029	2.1168e-05
	HHO	− 5.0877	− 5.0877	1.5811e-09

Table 3. Optimization results.

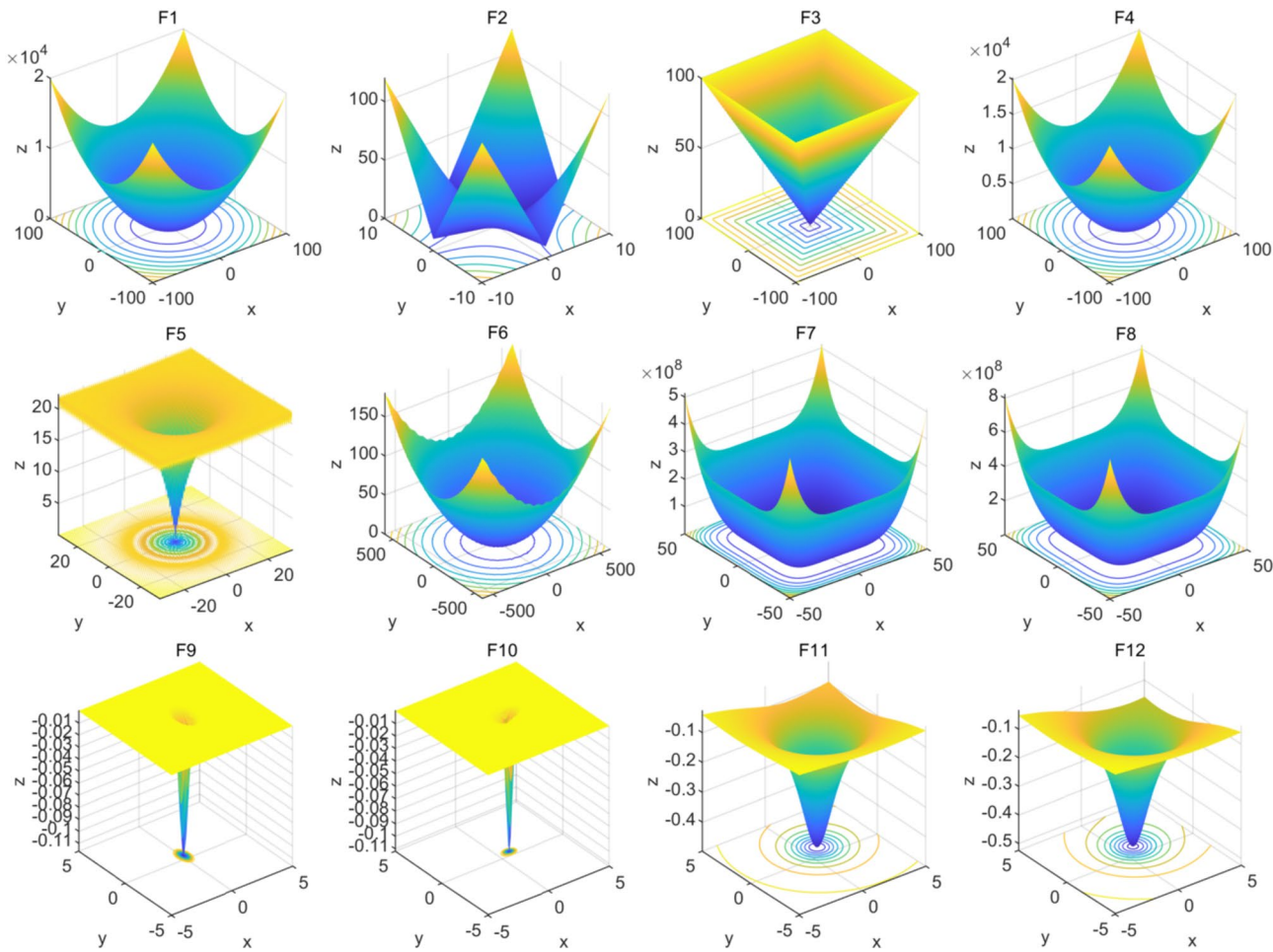


Figure 1. 2-D representation of benchmark functions.

deviation (std) values for each algorithm, providing insights into their optimization performance and stability. Table 3 presents the comparative results of each algorithm’s effectiveness in optimization. Additionally, Fig. 1 illustrates a 2-D representation of the function.

As shown in Table 3, GQPSO outperforms other algorithms in the optimization search of single-peak functions, particularly in consistently finding the optimal values for F1–F3. Despite showing slightly inferior performance to GWO for F4, it still surpasses QPSO and PSO. This observation also suggests that GQPSO possesses a robust global optimization search capability. When applied to multi-peak functions and fixed-dimension multi-peak functions, GQPSO also demonstrates superior performance than the other algorithms. In summary, GQPSO exhibits superior convergence speed and accuracy when dealing with single-peak functions, as well as robust resistance to local extrema in the context of multi-peak functions. Consequently, the incorporation of Lévy flight and Gaussian perturbation substantially enhances the algorithm’s stability and search capabilities.

Propose protocol

In QPSOFL, *n* homogeneous nodes with unique ID are grouped into *m* clusters. Each cluster has only one CH, and each CM belongs to only one cluster. The special node BS can be deployed within or outside the network, and once deployed, neither the BS nor the nodes can move. The proposed protocol nominates the most appropriate nodes as CHs by using quantum particle swarm optimization algorithm and selects the optimal relay CHs based

on fuzzy logic system to improve the network energy efficiency and maximize the network lifetime. The working of QPSOFL mainly consists of two phases, namely CH selection and relay finding.

CH selection

Selecting the optimal CHs from the nodes by using QPSO is one of the objectives of QPSOFL so as to form energy efficient clusters. At first, a population comprising N_p particles is initialized by using Sobol sequence with bounds in $[1, n]$. Each particle represents the set of nodes' ID that are candidates for the CHs, which can be express as $P_i = (q_{i,1}, q_{i,2}, \dots, q_{i,m})$ where each component $q_{i,j} (1 \leq j \leq m)$ denotes CH j in particle i . Note that the non-integer values are converted to the nearest integer in the implementation. Moreover, only the nodes whose energy is higher than the average are eligible to be candidate CHs for fast convergence.

Next, the particles are evaluated by a fitness function which considers energy and distance simultaneously. High residual energy is the first significant factor to consider for CH selection because CH consumes additional energy for data aggregation and forwarding compared with CM. Besides, the average intra- and inter-cluster distance is another importance factor to be considered, the smaller the distance, the lower the network energy consumption and the better the energy efficiency. Thus, the fitness function can be formulated in Eq. (8).

$$f = \alpha \times \frac{1}{\sum_{i=1}^m E_{res}(CH_i)} + (1 - \alpha) \times \frac{\sum_{i=1}^m \left(\frac{\sum_{j=1}^k d(s_j, CH_i)}{k} + d(CH_i, BS) \right)}{m} \quad (8)$$

where $E_{res}(CH_i)$ denotes the residual energy of CH i , $d(s_j, CH_i)$ and $d(CH_i, BS)$ indicate the distances from CM j to CH i , and from CH i to the BS, respectively. k is the number of CMs in the corresponding cluster, and $\alpha = 0.6$ is the co-efficiency ranging in $[0, 1]$ used to regulate the factors of energy and distance. In order to obtain the residual energy of the nodes, energy model as the same in^{9,36} is utilized to calculate the energy consumption of data transmission with k bits over distance d between any two nodes i and j , and the energy consumption for data transmitting and receiving is expressed in Eqs. (9) and (10), respectively.

$$E_{tr}(k, d) = \begin{cases} k * E_{elec} + k * \varepsilon_{fs} * d^2, & d < d_0 \\ k * E_{elec} + k * \varepsilon_{mp} * d^4, & d \geq d_0 \end{cases} \quad (9)$$

$$E_{rc}(k) = k * E_{elec} \quad (10)$$

where E_{elec} represents the electronics energy consumption used for transmitting or receiving 1-bit data. ε_{fs} and ε_{mp} are the amplifier coefficients used for free space and multi-path model, and $d_0 = \sqrt{\varepsilon_{fs}/\varepsilon_{mp}}$ is the distance threshold for the model determination. Besides, the energy consumed for k -bit data aggregation can be calculated by using Eq. (11).

$$E_{DA} = k * E_{pDb} \quad (11)$$

E_{pDb} denotes the energy dissipated for fusing 1-bit data. Thereafter, the local best and global best can be obtained, and the new state of each particle in the next iteration also can be calculated by using Eq. (4). The process continues until the optimal solution is reached or the stopping condition is satisfied. The pseudo-code for CH selection in QPSOFL is given in Algorithm 1 below to demonstrate the steps used by QPSO technique for the calculation of the positions of the particles.

Input: n, m, Itr_{max}, N_p

Output: m IDs of CHs

begin

for i to N_p do

Generate the initial population using Eq. (3), the dimension of particles is m , and $pBest_i = P_i$

end for

Evaluate the quality of the particles using Eq. (8), and the position of particle $P_j (1 \leq j \leq N_p)$

with the maximum fitness is assigned to $gBest$

for $i=1$ to Itr_{max} do

for $j=1$ to N_p do

Update the state of P_j using Eq. (4)

Evaluate the quality of the particles using Eq. (8), and update the $pBest_j^i$ and $gBest$

end for

end for

end

Algorithm 1. Pseudo-code for the CH selection using QPSO

Finally, the particle with maximum fitness value becomes the global best $gBest = (q_{best,1}, q_{best,2}, \dots, q_{best,m})$. Once the CHs are selected, each CH sends an advertisement message to the nodes in its range announcing its CH identity, and the normal nodes join the nearest CH to form clusters. Furthermore, TDMA schedule mechanism is adopted by each CH to complete intra-cluster data transmission. That is to say each CM sends its sensed data to the CH in its assigned slot to avoid inference and collision, and the CH collects and aggregates the data from its CMs. Subsequently, the aggregated data is transmitted to the relay of the CH and till to the BS in the end.

Relay finding

In QPSOFL, a Mamdani fuzzy inference system is introduced to find optimal relay for each CH so as to decrease and balance the energy consumption of the CHs by creating a best path from any source CH to the BS. The fuzzy inference system consists of four components: fuzzifier, accepting the normalized crisp inputs and converting them to linguistic variables; fuzzy rule base, obtaining a set of if-then rules used for decision making from expert experience; inference engine, reasoning using the fuzzy rules to obtain the fuzzy output; defuzzifier, converting the fuzzy output into crisp values *Prob*. Residual energy $E_{res}(CH_i)$, energy deviation $E_{dev}(CH_i)$, and relay distance $D_{ry}(CH_i)$ are considered as inputs to the FIS for relay finding, then the framework of the FIS can be depicted in Fig. 2.

The fuzzy inference system generates probabilities for corresponding candidate relays of each CH based on the residual energy, energy deviation, and difference of relay distance. The input parameters play a critical role in decision making for relay finding, which are normalized to the same domains ranging in $[0, 1]$. After that, the membership function based on the expert experience is utilized to perform fuzzification, which are described as follows.

Residual energy $E_{res}(CH_i)$ denotes the remaining energy of candidate relay i of CH j , and it should be noted that only the neighbor CH i of CH j with residual energy higher than the average can become candidate relay. The relay needs more energy to forward data to the BS than the normal CHs, thus the more residual energy, the more likely the candidate relay is to become the final relay. Due to its importance for relay finding, three

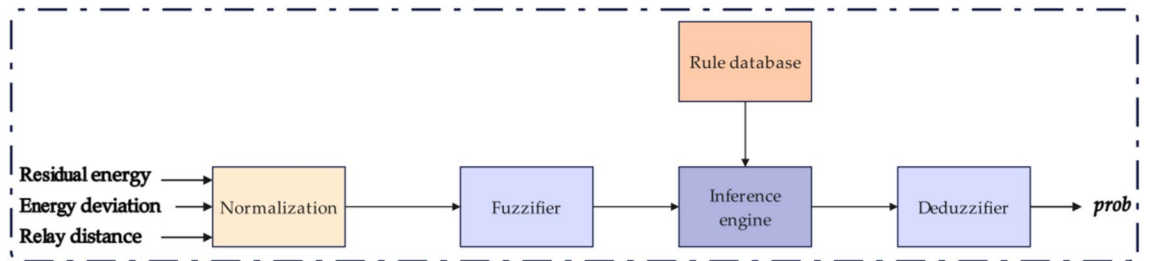


Figure 2. Fuzzy logic model in QPSOFL.

triangular and two trapezoidal membership functions are considered for residual energy, namely *very less*, *less*, *normal*, *much*, *very much*, which is shown in Fig. 3.

Energy deviation $E_{dev}(CH_i)$ denotes the deviation of energy consumption of candidate CH i , which indicates the consistency of energy consumption among the candidate CH i and its neighbor CHs. The smaller the standard deviation, the better the consistency of the energy consumption of these CHs, the better their balanced energy consumption, and even the less likely they are to cause hot spot problem. The energy deviation can be calculated by using Eq. (12).

$$E_{dev}(CH_i) = \left| E_{ec}(CH_i) - \frac{E_{ec}(CH_i) + \sum_{j=1}^k E_{ec}(CH_j)}{k+1} \right| \quad (12)$$

k is the number of neighbor CH located in the range of CH i , $E_{ec}(CH_i)$ denotes the energy that has been consumed by CH i . Three triangular and two trapezoidal membership functions are also considered for energy deviation, namely *very low*, *low*, *normal*, *high*, *very high* as shown in Fig. 4.

Relay distance $D_{ry}(CH_i)$ represents the sum of distances d_r and d_p as shown in Fig. 5. d_r is the distance from the candidate CH j to the straight line between CH i and the BS. d_p is the distance from candidate relay CH j to the crossover point CP_i between the communication circle of CH i and the straight line between CH i and the BS. The combination of the two distances represent not only energy saving but also fast data forwarding. That is to say, the smaller the relay distance, the more energy is saved, and the faster the data is forwarded to the BS.

Due to its critical influence for relay finding, three triangular and two trapezoidal membership functions are considered for relay distance, too, namely *very near*, *near*, *normal*, *far*, *very far*, which is shown in Fig. 6.

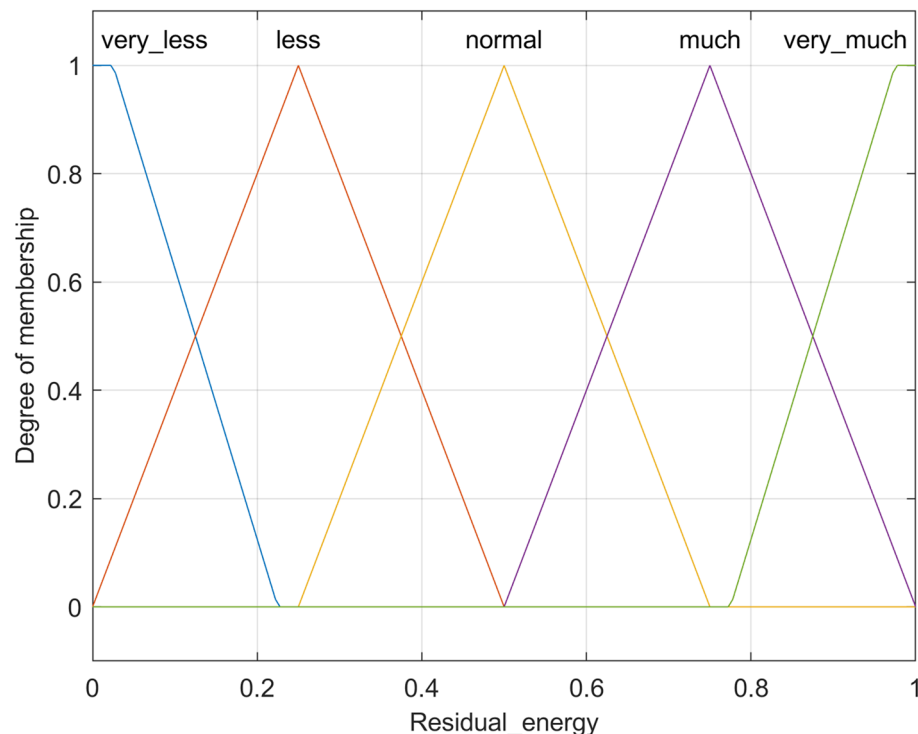


Figure 3. Membership function for residual energy.

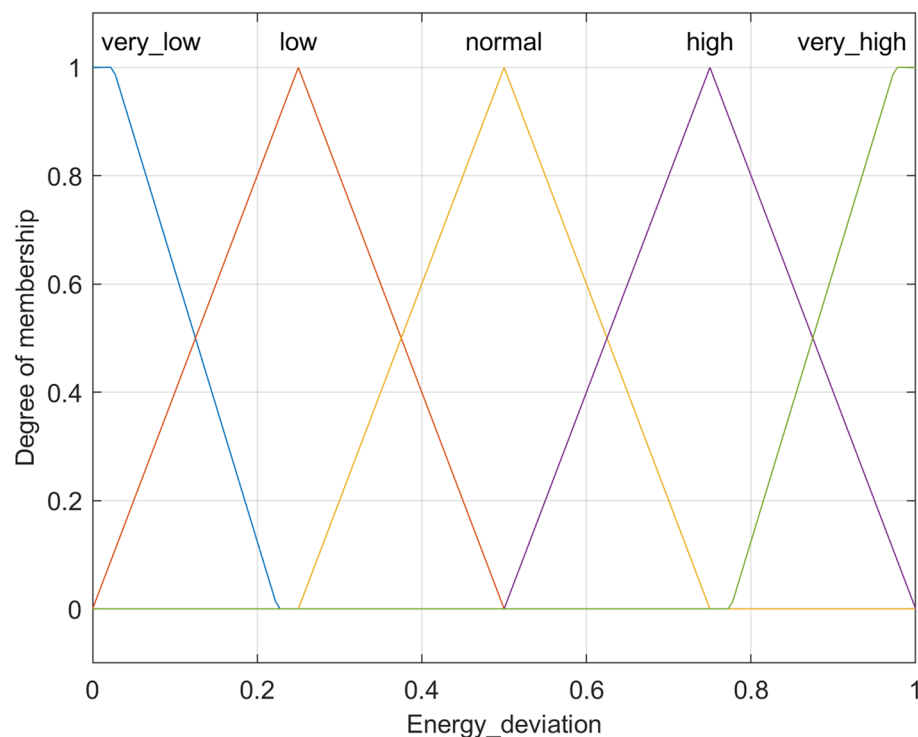


Figure 4. Membership function for energy deviation.

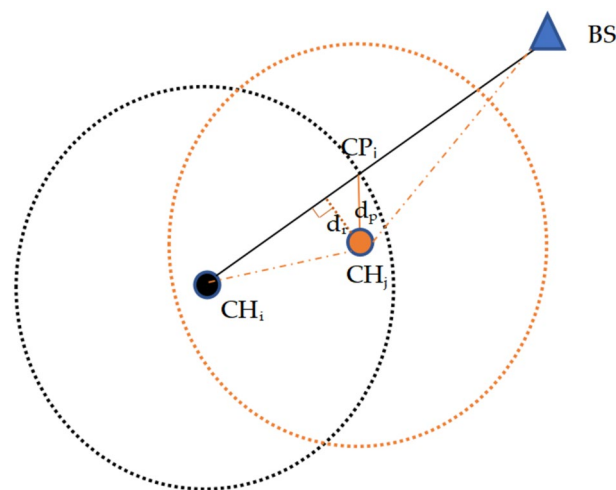


Figure 5. Relay distance considering energy saving and fast data forwarding.

After fuzzification of the input variables, 125 fuzzy rules based on expert experience are used for inference engine to obtain the fuzzy output, which is given in Table 4.

Thereafter, the fuzzy output is converted to the crisp value *Prob* by using CoA^{21,45}, who uses 5 triangular and 2 trapezoidal membership functions *very low*, *low*, *quite low*, *normal*, *quite high*, *high*, *very high*, as shown in Fig. 7.

When a candidate relay has a greater value *Prob*, it will have a greater probability of becoming a final relay. Once the relays are determined, the optimal path for data transmission from a source node to the BS is also decided. Each CM sends its sensing data to its CH at the allocated slot, and the CH aggregates all the data collected from its CMs, and then conveys the data to its relay, and so on until the data reaches the BS.

Time complexity

The time complexity of the QPSOFL protocol involves evaluating its main operational phases: cluster formation, route finding, and cluster and route maintenance. During cluster formation, QPSOFL employs Quantum

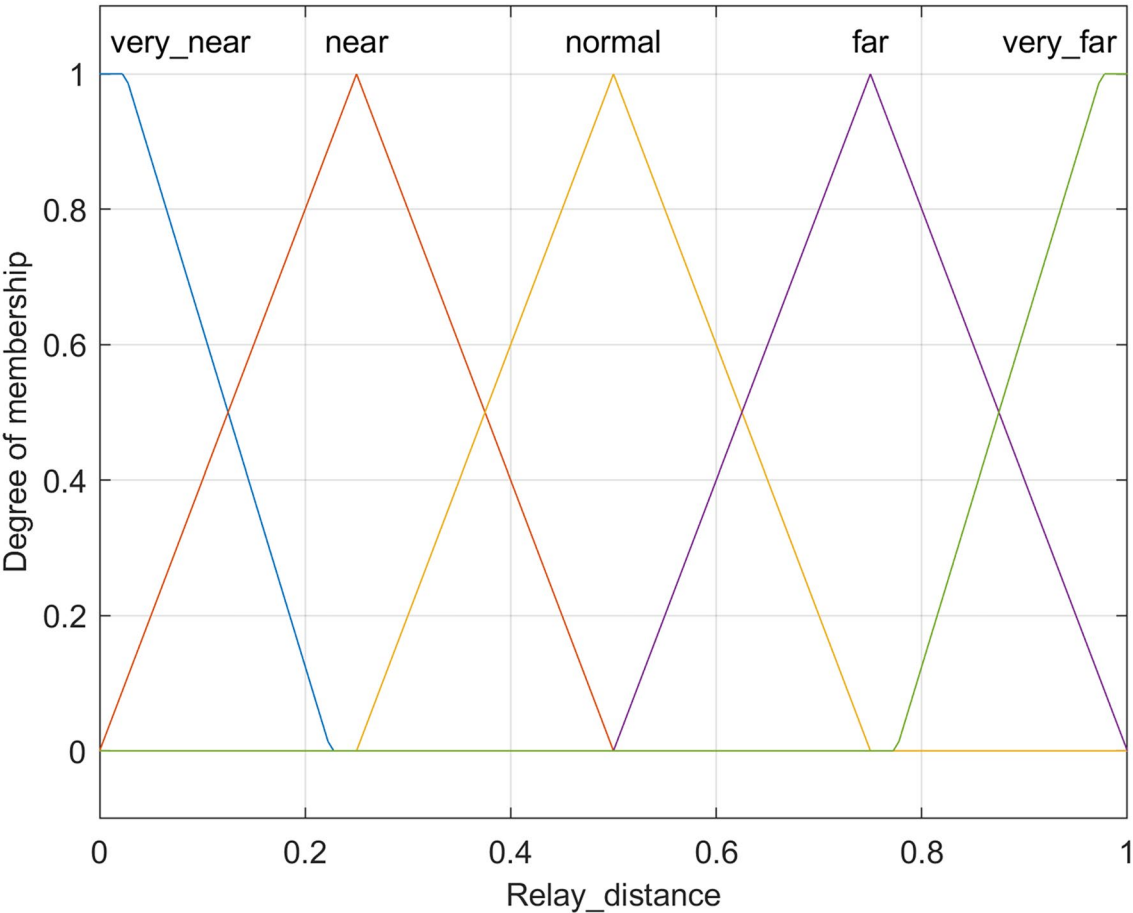


Figure 6. Membership function for relay distance.

Input variables	Output			
	Residual energy	Energy deviation	Relay distance	prob
1	Very less	Very high	Very far	Very low
2	Very less	Very high	far	Very low
3	Very less	Very high	normal	low
4	Very less	Very high	near	low
5	Very less	Very high	Very near	Quite low
...	...	...	...	...
125	Very much	Very low	Very near	Very high

Table 4. Fuzzy rules.

Particle Swarm Optimization, characterized by a time complexity of $O(n \times N_p \times \text{maxIter})$, where n denotes the number of sensor nodes, maxIter represents the maximum iterations, and N_p denotes the population size utilized in the optimization process. Subsequently, in route finding, QPSOFL integrates fuzzy logic to determine optimal relays for each CH. This stage is associated with a time complexity of $O(n_{CH} \times n_{rules})$, where n_{CH} stands for the number of CHs, n_{rules} represents the size of fuzzy rules. Finally, for cluster and route maintenance, local cluster updates require minimal messaging, specifically $O(n_{CH})$, while from the BS, broadcasting involves one message and forwarding entails a maximum of $O(n_{CH} - 1)$ messages. Therefore, the time complexity of QPSOFL is $O(n \times N_p \times \text{maxIter} + (n_{CH} + 1) \times n_{rules})$. In addition, n_p , n_{CH} , maxIter and n_{rules} are much less than n .

Human and animal rights

This article does not contain any studies with human participants or animals performed by any of the authors. No violation of Human and animal rights is involved.

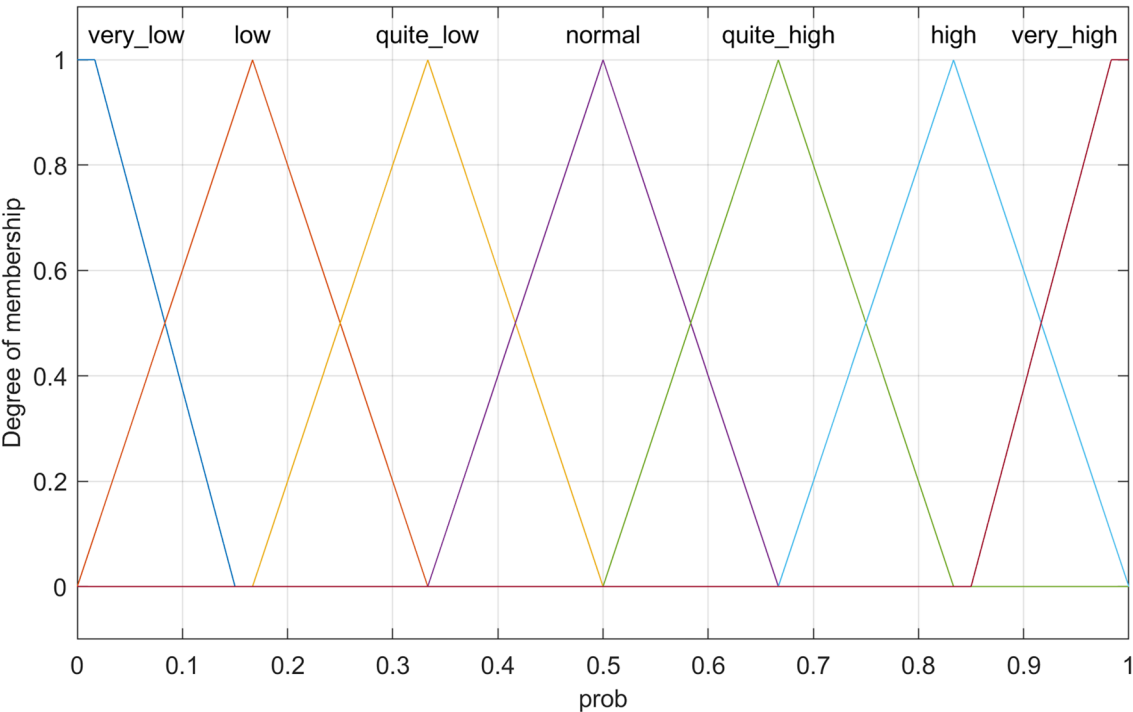


Figure 7. Membership function for output *Prob*.

Simulation results

The experiments were conducted on a computer running the Windows 10 operating system, equipped with an AMD Ryzen 5 3500X 6-Core processor, 16GB of RAM, and a 500GB SSD. The MATLAB R2022a platform is used to simulate and test the performance of the proposed QPSOFL. To ensure the robustness of the results, we repeated the experiments 50 times and calculated the average to draw the conclusions.

Simulation settings

We utilized identical simulation parameters as the majority of protocols, and the details of the network and radio energy models, which can be found in Table 5^{31,32}. In this study, as depicted in Fig. 8, we evaluate the network lifetime, throughput, energy consumption and scalability of the proposed protocol in the three scenarios (Fig. 8a–c). This evaluation involves varying the BS’s location and testing the scalability of QPSOFL by altering the network area’s size.

Parameters	Values
Number of nodes	100
Network area	200m*200m
Initial energy	0.5 J
E _{elec}	50 (nJ/bit)
E ₀	5(nJ/bit)
ε _{fs}	10 (pJ/bit/m ²)
ε _{mp}	0.0013 (pJ/bit/m ⁴)
d ₀	87.7 m
Data packet size	4000bits
Control packet size	200bits
BS location	x = 100, y = 100 x = 100, y = 200 x = 100, y = 250

Table 5. Simulation parameters.

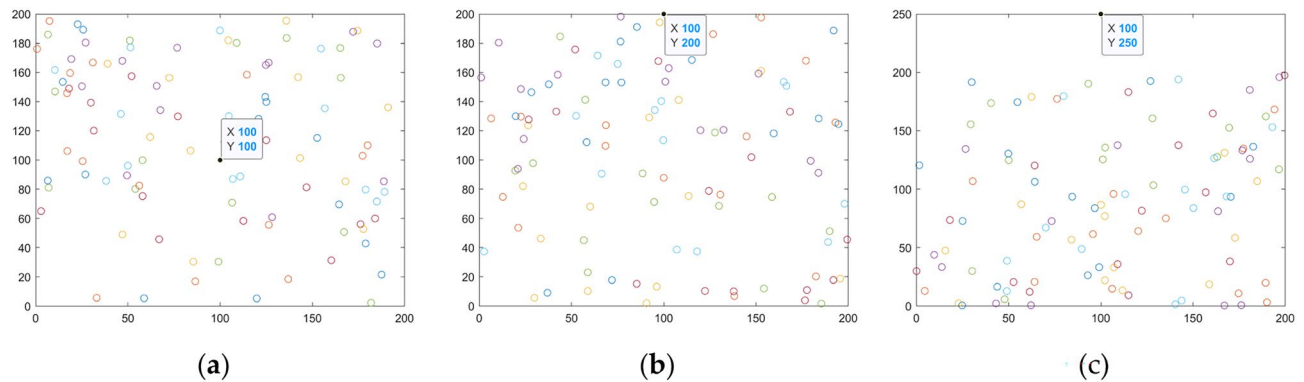


Figure 8. Network test environment. **a** BS coordinates is (100,100); **b** BS coordinates is (100,200); **c** BS coordinates is (100,250).

State-of-the-art algorithms for comparison

To rigorously validate the performance of the proposed QPSOFL protocol, we conducted a comparative analysis. This analysis involved a comparison with protocols utilizing fuzzy systems, as well as protocols that combine various meta-heuristic algorithms and fuzzy systems. These comparisons were conducted within the context of bench-marking, and the protocols involved in this assessment include approaches such as E-FUCA³⁶, IHHO-F⁴³, F-GWO⁴⁴, and FLPSOC⁴⁶.

Simulation result and discussion

Network lifetime

In WSNs, network lifetime serves as a critical performance metric, representing the duration for which the entire network can operate effectively. This duration indicates how long the network can function before its sensor nodes deplete their energy. Given that wireless sensor nodes are typically battery-powered, with limited lifespans, it is imperative to manage energy efficiently in order to prolong the network lifetime. The network's lifespan is closely linked to the count of operational nodes, and its performance is typically assessed through metrics such as FND (First Node Dead), HND (Half of the Nodes Dead), and LND (Last Node Dead). The results of the comparison between QPSOFL and E-FUCA, IHHO-F, F-GWO, and FLPSOC are presented in Fig. 9 and Table 6, respectively.

As depicted in Fig. 9, despite the varying positions of the BS, it remains evident that QPSOFL outperforms the other algorithms in all the cases. For IHHO-F, despite considering factors like energy, distance, and the number of neighboring nodes in the selection of the optimal cluster head set, excessive data transmission loads can result in the early depletion of SCH nodes, consequently impacting the network's overall lifetime. In F-GWO, an energy-efficient opportunistic routing method is employed during the data transmission phase. However, a drawback is the absence of distance consideration during clustering, potentially resulting in the selection of CHs located far from the BS, leading to increased energy consumption over extended transmission distances. E-FUCA utilizes a fuzzy logic system to make decisions during the cluster routing phase, however, relying solely on the output of a fuzzy system cannot adapt to the dynamic environment during data transmission within a wireless sensor network. In contrast to FLPSOC, the proposed QPSOFL, first utilizes the QPSO algorithm to select the optimal CHs, and subsequently employs fuzzy logic to find the relay nodes. The number of rounds in which nodes experience depletion can be observed individually in Table 4. Overall, QPSOFL considers factors such as residual energy, distance, standard deviation of energy, and leverages the distributional properties of the fuzzy logic system to maintain a balance among these parameters, ultimately leading to an extended network lifetime. In summary, QPSOFL outperforms the other algorithms in extending network lifetime with a performance

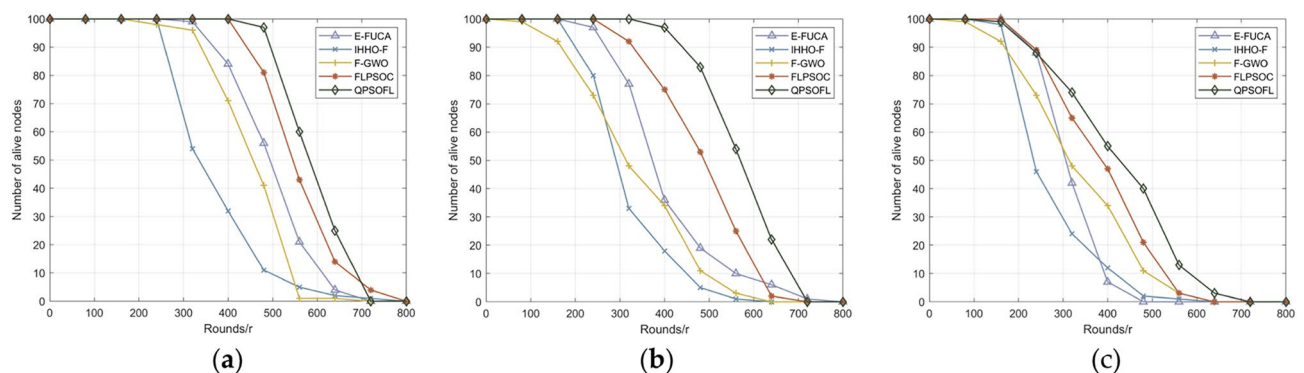


Figure 9. Comparison of the number of alive nodes. **a** BS coordinates is (100,100); **b** BS coordinates is (100,200); **c** BS coordinates is (100,250).

	QPSOFL	FLPSOC	F-GWO	IHHO-F	E-FUCA	
Scenario #1	FND	469	427	203	250	321
	HND	583	544	455	331	495
	LND	713	759	712	754	682
Scenario #2	FND	345	279	70	185	222
	HND	567	488	311	269	377
	LND	716	659	611	637	789
Scenario #3	FND	146	204	56	151	178
	HND	419	376	308	235	307
	LND	647	604	604	601	448

Table 6. Comparison of FND, HND and LND.

advantage of 1.92%, 12.40%, and 2.22% over FLPSOC; 22.33%, 39.06%, and 20.06% over F-GWO; and 24.31%, 32.98%, and 18.49% over IHHO-F, respectively. Furthermore, it surpasses E-FUCA by 15.07%, 14.74%, and 22.95%, respectively.

Network throughput

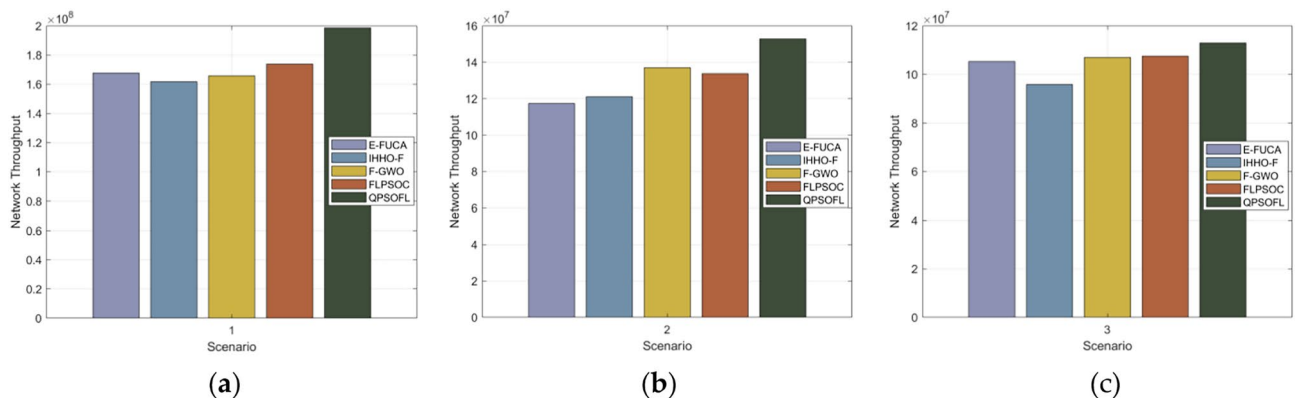
Throughput constitutes a crucial performance metric employed to characterize the volume of data or information that a network can effectively transmit within a given time frame. Typically, Throughput serves as a standard measure for assessing a network's data transfer capacity and overall performance. Enhanced throughput can significantly enhance network performance, diminish transmission latency, and broaden the range of supported application scenarios, including monitoring, control, and data transfer. Figure 10 illustrates the test results of network throughput.

As depicted in Fig. 10, it is evident that QPSOFL surpasses E-FUCA, IHHO-F, F-GWO, and FLPSOC in network throughput across all three scenarios. This is attributed to the data received by the BS, which is contingent on energy consumption and the network's lifecycle. QPSOFL enhances network throughput over E-FUCA by 15.59%, 23.15%, and 6.76%; surpasses IHHO-F by 18.56%, 20.87%, and 15.22%; outperforms F-GWO by 16.48%, 10.37%, and 5.31%; and exceeds FLPSOC by 12.46%, 12.51%, and 4.81%, respectively. In the case of IHHO-F, low network throughput can be attributed to high energy consumption and a shortened network lifespan resulting from excessive SCH loading. E-FUCA can be optimized to create an ideal cluster utilization scheme through the use of a fuzzy logic system, thereby extending the network's lifespan and enhancing network throughput. Similarly, FLPSOC utilizes a fuzzy logic system to establish appropriate clustering and relay node selection through PSO. Consequently, the overall network throughput performance surpasses that of E-FUCA. Regarding F-GWO, the throughput performance is likewise superior, primarily due to the increased data transmission to the BS facilitated by an energy-efficient opportunistic routing scheme during the data transmission phase.

Energy consumption

Energy consumption is a critical performance metric that quantifies the battery energy utilized by wireless sensor nodes during their tasks and communication. Wireless sensor nodes are typically battery-powered, making energy consumption a pivotal factor. Figure 11 displays the results of comparing QPSOFL with E-FUCA, IHHO-F, F-GWO, and FLPSOC.

As depicted in Fig. 11, network energy consumption rises with the number of rounds, yet the network energy profile of QPSOFL remains notably lower compared to other protocols. In IHHO-F, the SCH nodes become

**Figure 10.** Comparison of the network throughput. **a** BS coordinates is (100,100); **b** BS coordinates is (100,200); **c** BS coordinates is (100,250).

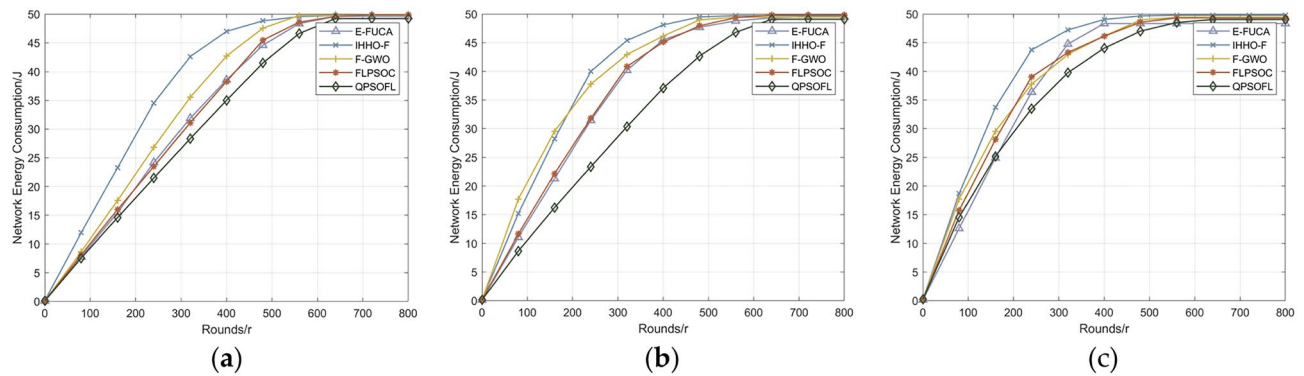


Figure 11. Comparison of the network energy consumption. **a** BS coordinates is (100,100); **b** BS coordinates is (100,200); **c** BS coordinates is (100,250).

overloaded, leading to increased network energy consumption. In the case of F-GWO, the CHs located far from the BS experience high energy consumption due to their distant proximity to the BS. In EFUCA, a fuzzy logic approach is employed to create unbalanced clusters, aiming to conserve intra-cluster energy consumption. Subsequently, a fuzzy logic system is utilized to determine the optimal routing path, reducing inter-cluster energy consumption. Consequently, the overall network energy consumption is typically lower than that of IHHO-F and F-GWO. In contrast, FLPSOC first chooses the optimal CHs based on the output of the fuzzy logic system and then employs the PSO algorithm to select the best relay node. This approach results in a reduction of energy consumption to some degree. QPSOFL outperforms IHHO-F by 38.51%, 45.38%, and 28.30%, surpasses F-GWO by 20.49%, 50.76%, and 19.49%, exceeds E-FUCA by 12.01%, 27.30%, and 1.88%, and outpaces FLPSOC by 9.18%, 29.23%, and 12.57%, respectively. These findings highlight the significant energy-efficiency improvements achieved by QPSOFL compared to other protocols.

Network scalability

Network scalability is of paramount importance in wireless sensor networks, commonly employed for extensive monitoring and data collection tasks spanning various applications. Clustering and routing protocols with strong scalability can adaptly manage network expansion and alterations, all while safeguarding network performance, energy efficiency, and data transmission quality. Figure 12 illustrates the network lifetime results of the suggested QPSOFL protocol within areas of 200m * 200m and 800m * 800m, with the BS positioned at the center of each area.

As depicted in Fig. 12, with the expansion of the network area, IHHO-F experiences premature degradation owing to node overloading during the routing phase, whereas F-GWO progressively demonstrates its superiority with the enlargement of the network area, attributed to the implementation of an energy-efficient routing scheme in the routing phase. Our QPSOFL protocol, as proposed, integrates energy consumption considerations in both the clustering and routing phases. It selects the most optimal scheme in each phase, leading to superior scalability when compared to other protocols.

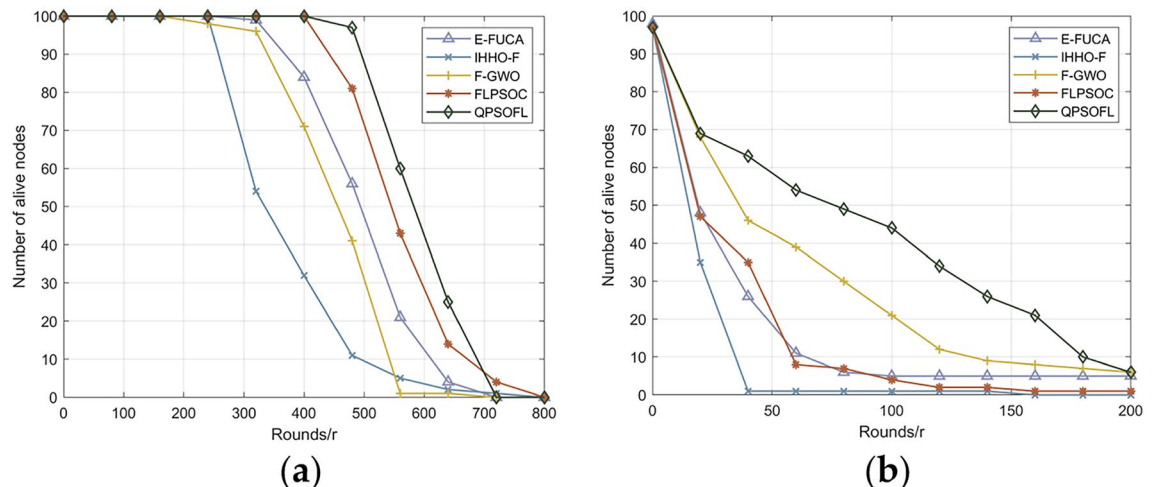


Figure 12. Comparison of the network energy consumption. **a** Area size is 200*200; **b** Area size is 800*800.

Result analysis

Simulation results indicate that the QPSOFL protocol, as proposed, exhibits substantial enhancements in key performance metrics, including network survival period, throughput, network energy consumption, and scalability, offering significant advantages. Upon a comprehensive examination of the simulation analysis, the specific factors contributing to this enhancement can be attributed to the following: The initial application of the enhanced QPSO algorithm to select the optimal set of CHs based on a carefully designed fitness function, resulting in CHs with both higher residual energy and closer proximity to the BS. This leads to a substantial reduction in energy consumption. Additionally, optimal relay nodes are chosen during the routing phase by employing a fuzzy logic system, where the input parameters of this system are meticulously designed to optimize the averaging of network energy load. These factors collectively contribute to the notable performance improvements observed in the QPSOFL protocol.

Conclusion

This paper introduces QPSOFL, a novel clustering and routing protocol aimed at simultaneously enhancing energy efficiency and extending network lifetime. The protocol integrates quantum particle optimization for cluster formation and fuzzy logic for relay finding, prioritizing both energy efficiency and balance. Experimental results demonstrate that QPSOFL significantly outperforms classical and current clustering and routing protocols such as E-FUCA, IHHO-F, F-GWO, and FLPSOC. QPSOFL effectively prolongs network lifetime, reduces energy consumption, balances energy distribution, and achieves higher throughput for data transmission to the base station. Moreover, QPSOFL exhibits superior scalability compared to its counterparts. In future research, we aim to enhance the QPSOFL protocol's adaptability to mobile and rechargeable nodes, thereby improving overall network performance and extending network lifetime. Key areas of focus include investigating mobility management strategies, integrating energy harvesting techniques, optimizing resource allocation dynamically, and strengthening security measures to ensure robust and reliable operation in diverse and challenging environments.

Code availability

Data will be provided by the corresponding author upon reasonable request by the reader.

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Author contributions

Conceptualization, H.H. and X.F.; methodology, C.W.; software, X.F.; validation, H.H., X.F. and C.W.; formal analysis, H.H.; investigation, X.F.; resources, C.W.; data curation, X.F.; writing—original draft preparation, H.H.; writing—review and editing, C.W.; visualization, X.F.; supervision, C.W.; project administration, H.H. and C.W.; funding acquisition, H.H. and C.W. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to C.W.

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